

SIMATS ENGINEERING

ASSESSMENT TOOL 2

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COURSE NAME: COMPUTER NETWORKS

COURSE CODE: CSA0703

COURSE OUTCOME COVERED

CO2: Configure IP addressing schemes, subnetting, and basic routing techniques using simulation tool, for efficient network design and topology. *(BL3)*

Assessment Tool Weightage: 30%

Annexure A

Scenario-Based

1. A newly established Smart Campus is deploying a secure network infrastructure for its academic and administrative departments. The network administrator has been allocated the IP address block 192.168.100.0/24. Based on the expected number of devices, the following subnet masks have already been assigned:

School of Computing → /25

School of Electronics → /26

School of Mechanical Engineering → /27

As the network engineer, design the subnet allocation by performing the following tasks:

- 1. Determine the network address and broadcast address for each department.**
- 2. Identify the valid host IP address range for each subnet.**
- 3. Calculate the number of usable host addresses available in each subnet.**
- 4. Determine the remaining unused IP address range within the given network block.**

Scenario

A newly established **Smart Campus** is deploying a secure network infrastructure for its academic and administrative departments. The network administrator has been allocated the IP address block:

Network Address: 192.168.100.0/24

The network is divided into three subnets based on the expected number of devices:

Department	Subnet Mask	CIDR Notation
School of Computing	255.255.255.128	/25
School of Electronics	255.255.255.192	/26
School of Mechanical Engineering	255.255.255.224	/27

Task 1: Determine the Network Address and Broadcast Address

Explanation

- The Network Address is the first IP address in each subnet.
- The Broadcast Address is the last IP address in each subnet.
- These two addresses are reserved and cannot be assigned to devices.

Department	Subnet	Network Address	Broadcast Address
School of Computing	/25	192.168.100.0	192.168.100.127
School of Electronics	/26	192.168.100.128	192.168.100.191
School of Mechanical Engineering	/27	192.168.100.192	192.168.100.223

Task 2: Identify the Valid Host IP Address Range

Explanation

The valid host addresses lie between the network address and the broadcast address.

- First Host = Network Address + 1
- Last Host = Broadcast Address – 1

Department	First Host IP	Last Host IP
School of Computing	192.168.100.1	192.168.100.126
School of Electronics	192.168.100.129	192.168.100.190
School of Mechanical Engineering	192.168.100.193	192.168.100.222

Task 3: Calculate the Number of Usable Host Addresses

Formula

$$\text{Usable Hosts} = 2^{(32 - \text{Prefix Length})} - 2$$

Why subtract 2?

- One address is reserved as the **Network Address**.
- One address is reserved as the **Broadcast Address**.

Calculation:

School of Computing (/25)

Host Bits:

$$32 - 25 = 7$$

Total Addresses:

$$2^7 = 128$$

Usable Hosts:

$$128 - 2 = 126$$

School of Electronics (/26)

Host Bits:

$$32 - 26 = 6$$

Total Addresses:

$$2^6 = 64$$

Usable Hosts:

$$64 - 2 = 62$$

School of Mechanical Engineering (/27)

Host Bits

$$32 - 27 = 5$$

Total Addresses

$$2^5 = 32$$

Usable Hosts

$$32 - 2 = 30$$

Summary

Department	Prefix	Total Addresses	Usable Hosts
School of Computing	/25	128	126
School of Electronics	/26	64	62
School of Mechanical Engineering	/27	32	30

Task 4: Determine the Remaining Unused IP Address Range

Allocated Subnets

Department	IP Address Range
Computing	192.168.100.0 – 192.168.100.127
Electronics	192.168.100.128 – 192.168.100.191
Mechanical	192.168.100.192 – 192.168.100.223

The original network is **192.168.100.0/24**, which contains IP addresses from:

192.168.100.0 – 192.168.100.255

After allocating the three subnets, the remaining addresses are:

Remaining Unused Range	Number of Addresses
192.168.100.224 – 192.168.100.255	32 Addresses

Conclusion

The allocated 192.168.100.0/24 network has been successfully divided into three subnets using Variable Length Subnet Masking (VLSM). Each department receives a subnet sized according to its requirements, ensuring efficient IP address utilization while leaving 192.168.100.224–192.168.100.255 available for future expansion. This design minimizes address wastage and supports scalable network growth.

2. A multinational software company is expanding its headquarters and has been assigned the private IP block 10.10.0.0/16. Different departments require different subnet sizes based on the number of employees. The network administrator has allocated the following subnet masks:

Software Development → /24

Quality Assurance → /25

Human Resources → /26

Executive Management → /27

As the network administrator, perform the following tasks:

- 1. Determine the network address and broadcast address for each department.**
- 2. Identify the valid range of host IP addresses.**
- 3. Calculate the number of usable hosts in each subnet.**
- 4. Identify the unused IP address range remaining within the allocated IP block.**

A multinational software company is expanding its headquarters and has been assigned the private IPv4 network:

Network Address: 10.10.0.0/16

To accommodate different departments based on the number of employees, the network administrator has allocated the following subnet masks:

Department	Subnet Mask	CIDR Notation
Software Development	255.255.255.0	/24
Quality Assurance	255.255.255.128	/25
Human Resources	255.255.255.192	/26
Executive Management	255.255.255.224	/27

Task 1: Determine the Network Address and Broadcast Address

Explanation

- The **Network Address** is the **first address** of the subnet.
- The **Broadcast Address** is the **last address** of the subnet.
- These addresses are reserved and cannot be assigned to hosts.

Department	Subnet	Network Address	Broadcast Address
Software Development	/24	10.10.0.0	10.10.0.255
Quality Assurance	/25	10.10.1.0	10.10.1.127
Human Resources	/26	10.10.1.128	10.10.1.191

Executive Management	/27	10.10.1.192	10.10.1.223
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Task 2: Identify the Valid Host IP Address Range

Explanation

The usable host IP addresses are between the **network address** and the **broadcast address**.

- **First Host = Network Address + 1**
- **Last Host = Broadcast Address – 1**

Department	First Host IP	Last Host IP
Software Development	10.10.0.1	10.10.0.254
Quality Assurance	10.10.1.1	10.10.1.126
Human Resources	10.10.1.129	10.10.1.190
Executive Management	10.10.1.193	10.10.1.222

Task 3: Calculate the Number of Usable Hosts

Formula

$$\text{Usable Hosts} = 2^{(32 - \text{Prefix Length})} - 2$$

Why subtract 2?

- One IP address is reserved as the **Network Address**.
- One IP address is reserved as the **Broadcast Address**.

Calculations

Software Development (/24)

$$\text{Host Bits: } 32 - 24 = 8$$

$$\text{Total Addresses: } 2^8 = 256$$

$$\text{Usable Hosts: } 256 - 2 = 254$$

Quality Assurance (/25)

$$\text{Host Bits: } 32 - 25 = 7$$

$$\text{Total Addresses: } 2^7 = 128$$

$$\text{Usable Hosts: } 128 - 2 = 126$$

Human Resources (/26)

Host Bits: $32 - 26 = 6$

Total Addresses: $2^6 = 64$

Usable Hosts: $64 - 2 = 62$

Executive Management (/27)

Host Bits: $32 - 27 = 5$

Total Addresses: $2^5 = 32$

Usable Hosts: $32 - 2 = 30$

Summary Table

Department	Prefix	Total Addresses	Usable Hosts
Software Development	/24	256	254
Quality Assurance	/25	128	126
Human Resources	/26	64	62
Executive Management	/27	32	30

Task 4: Identify the Remaining Unused IP Address Range

Allocated Subnets

Department	Allocated IP Range
Software Development	10.10.0.0 – 10.10.0.255
Quality Assurance	10.10.1.0 – 10.10.1.127
Human Resources	10.10.1.128 – 10.10.1.191
Executive Management	10.10.1.192 – 10.10.1.223

The original network **10.10.0.0/16** contains addresses from:

10.10.0.0 – 10.10.255.255

After allocating the required subnets, the remaining available IP address range is:

Remaining Unused IP Range	Available Addresses
10.10.1.224 – 10.10.255.255	Remaining addresses for future expansion

Explanation: Since the allocated subnets occupy addresses from **10.10.0.0** to **10.10.1.223**, all addresses from **10.10.1.224** up to **10.10.255.255** remain unused and can be allocated to future departments or network expansion.

Conclusion

The 10.10.0.0/16 network has been successfully subnetted using Variable Length Subnet Masking (VLSM) to meet the different host requirements of each department. Each department receives an appropriately sized subnet, ensuring efficient IP address utilization while leaving a large portion of the address space (10.10.1.224 – 10.10.255.255) available for future organizational growth and additional departments.

3.A smart hospital is migrating its network infrastructure from IPv4 to IPv6 to support thousands of connected medical devices and IoT sensors. The hospital has been assigned the IPv6 network:

2001:0db8:abcd:0000:0000:0000:0000/64

As the network engineer responsible for the migration, perform the following:

- 1. Convert the given IPv6 address into its compressed notation.**
- 2. Expand the compressed address back into its full hexadecimal representation.**
- 3. Identify the network prefix using the /64 prefix length.**
- 4. Calculate the total number of host addresses supported within the subnet.**
- 5. the total number of possible host addresses available within this network.**

IPv6 Address Structure

2001 : 0db8 : abcd : 0000 : 0000 : 0000 : 0000

| <----- Network Prefix (/64) -----> | <----- Host ID
(/64) -----> |

Explanation:

- An IPv6 address contains 128 bits.
- Each block (hextet) contains 16 bits.
- A /64 prefix means:
 - The first 64 bits identify the network.

- The last 64 bits identify the host/interface.

Task 1: Convert the IPv6 Address into Compressed Notation

Given IPv6 Address

2001:0db8:abcd:0000:0000:0000:0000/64

Compression Rules

1. Remove leading zeros from each block.
2. Replace one continuous sequence of zero blocks with ::.

Compressed IPv6 Address

2001:db8:abcd::/64

Explanation

- 0db8 becomes db8.
- The five consecutive 0000 blocks are replaced with ::.
- The :: notation can be used **only once** in an IPv6 address.

Task 2: Expand the Compressed Address

Compressed Address

2001:db8:abcd::/64

Expanded Address

2001:0db8:abcd:0000:0000:0000:0000/64

Explanation

When expanding:

- Every omitted block is restored as 0000.
- Each hextet must contain exactly four hexadecimal digits.

Task 3: Identify the Network Prefix Using /64

Explanation

An IPv6 address has 128 bits.

A /64 prefix means:

- Network Portion = First 64 bits
- Host Portion = Last 64 bits

Network Prefix

2001:0db8:abcd:0000::/64

Network and Host Division

Portion	Value
Network Prefix (64 bits)	2001:0db8:abcd:0000
Host Identifier (64 bits)	0000:0000:0000:0000

Task 4: Calculate the Total Number of Host Addresses

Formula

$$\text{Number of Host Addresses} = 2^{(128 - \text{Prefix Length})}$$

Since the prefix length is /64,

Host Bits: $128 - 64 = 64$

Therefore,

$$2^{64} = 18,446,744,073,709,551,616$$

Answer

Total host addresses supported = 18,446,744,073,709,551,616

Task 5: Total Number of Possible Host Addresses

IPv6 does not reserve separate network and broadcast addresses as IPv4 does.

Therefore, the total number of possible addresses in this /64 subnet is:

$$2^{64} = 18,446,744,073,709,551,616$$

Total possible host addresses = 18,446,744,073,709,551,616

Summary Table

Task	Result
Original IPv6 Address	2001:0db8:abcd:0000:0000:0000:0000:0000/64
Compressed Address	2001:db8:abcd::/64
Expanded Address	2001:0db8:abcd:0000:0000:0000:0000:0000/64
Network Prefix (/64)	2001:0db8:abcd:0000::/64
Network Portion	2001:0db8:abcd:0000
Host Portion	0000:0000:0000:0000

Host Bits	64 bits
Total Host Addresses	$2^{64} = 18,446,744,073,709,551,616$
Total Possible Host Addresses	18,446,744,073,709,551,616

Conclusion

The hospital's IPv6 network 2001:0db8:abcd:0000::/64 uses a 64-bit network prefix and a 64-bit host identifier, providing 2^{64} (18,446,744,073,709,551,616) unique addresses. This enormous address space is ideal for supporting thousands of connected medical devices, IoT sensors, patient monitoring systems, and future network expansion without the address limitations of IPv4.

4. An Internet Service Provider (ISP) connects multiple customer networks through a core router. The router maintains the following routing table entries:

192.168.1.0/24

192.168.2.0/24

192.168.3.0/24

A packet arrives at the router with the destination IP address:

192.168.2.45

As the routing engineer, determine:

- 1. The destination subnet to which the packet belongs.**
- 2. The matching routing table entry.**
- 3. The next-hop network/interface used to forward the packet.**
- 4. The default route that should be used if the destination does not match any routing table entry.**

Scenario

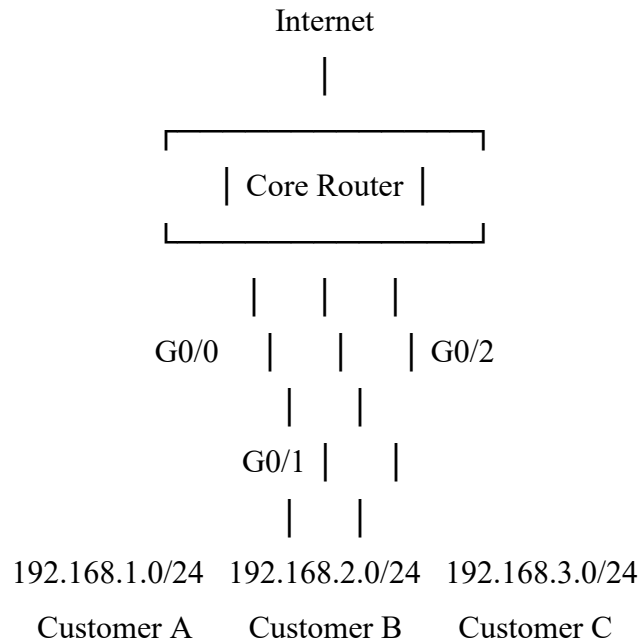
An Internet Service Provider (ISP) connects multiple customer networks through a **core router**. The router maintains the following routing table entries:

Destination Network	Subnet Mask	CIDR
192.168.1.0	255.255.255.0	/24
192.168.2.0	255.255.255.0	/24
192.168.3.0	255.255.255.0	/24

A packet arrives at the router with the destination IP address:

Destination IP: 192.168.2.45

Network Topology



Destination Packet → 192.168.2.45

Routing Table

Destination Network	Prefix	Interface
192.168.1.0	/24	GigabitEthernet0/0
192.168.2.0	/24	GigabitEthernet0/1
192.168.3.0	/24	GigabitEthernet0/2

Task 1: Determine the Destination Subnet

Explanation

The destination IP address is:

192.168.2.45

For a /24 subnet:

- Network Address = 192.168.2.0
- Broadcast Address = 192.168.2.255
- Valid Host Range = 192.168.2.1 – 192.168.2.254

Since 192.168.2.45 falls within this range, it belongs to the subnet:

192.168.2.0/24

Answer

Destination IP	Destination Subnet
192.168.2.45	192.168.2.0/24

Task 2: Identify the Matching Routing Table Entry

Explanation

The router compares the destination IP with each routing table entry.

Routing Table Entry	Match
192.168.1.0/24	✗ No
192.168.2.0/24	☑ Yes
192.168.3.0/24	✗ No

The destination IP matches: **192.168.2.0/24**

Task 3: Determine the Next-Hop Network / Interface

Explanation

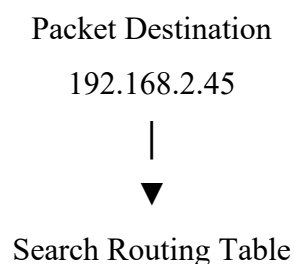
Since the destination network is **192.168.2.0/24**, the router forwards the packet through the interface connected to that network.

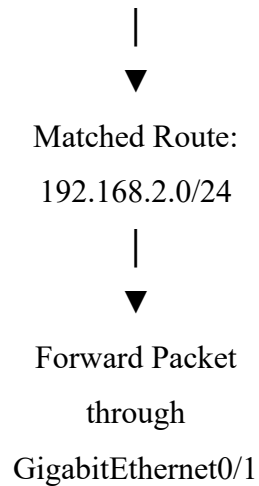
Assume the routing table is configured as follows:

Destination Network	Interface
192.168.1.0/24	GigabitEthernet0/0
192.168.2.0/24	GigabitEthernet0/1
192.168.3.0/24	GigabitEthernet0/2

Therefore, the packet is forwarded through: **GigabitEthernet0/1**

Packet Forwarding Process





Answer

Destination Network	Outgoing Interface
192.168.2.0/24	GigabitEthernet0/1

Task 4: Identify the Default Route

Explanation

If the destination IP address does not match any entry in the routing table, the router forwards the packet using the default route.

The default route is represented as:

0.0.0.0/0

Example configuration:

```
ip route 0.0.0.0 0.0.0.0 <Next-Hop-IP>
```

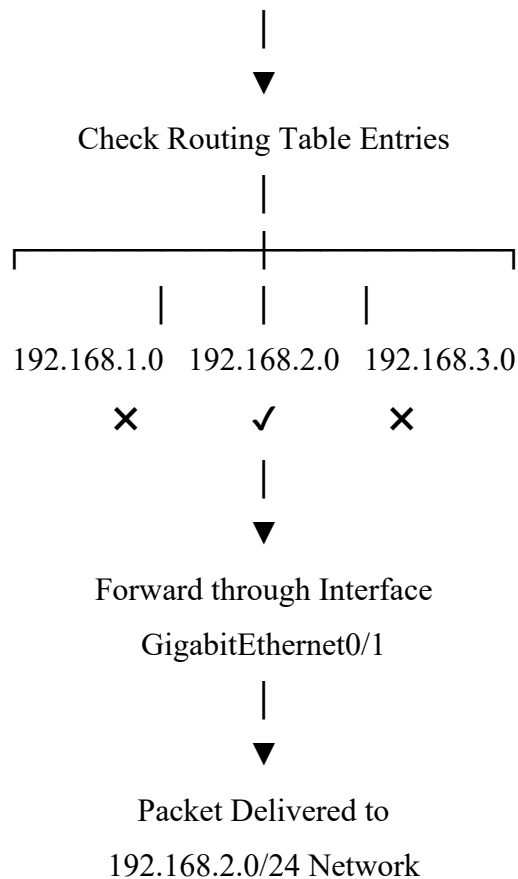
The Next-Hop-IP is the IP address of another router (such as the ISP gateway) that will receive packets destined for unknown networks.

Answer

Condition	Route Used
No matching destination network	0.0.0.0/0 (Default Route)

Routing Decision Flow

Packet Arrives
Destination IP:
192.168.2.45



Conclusion

The destination IP address 192.168.2.45 belongs to the 192.168.2.0/24 subnet. The router matches this destination with the corresponding routing table entry and forwards the packet through GigabitEthernet0/1. If no matching route is found, the router uses the default route (0.0.0.0/0) to send the packet to the next-hop router, ensuring communication with unknown or external networks.

5. A company establishes two office departments connected through a Layer-3 router. The allocated subnet information is as follows:

Administration LAN → 192.168.10.0/26

Finance LAN → 192.168.10.64/26

The router interfaces are configured with:

Administration Gateway → 192.168.10.1

Finance Gateway → 192.168.10.65

As the network administrator, perform the following tasks:

- 1. Determine the network address and broadcast address for each LAN.
- 2. Identify the valid host IP address range.
- 3. Suggest suitable IP addresses for three computers in each LAN.
- 4. Identify the default gateway that hosts in each LAN should use for communication.

Scenario

A company has established two office departments connected through a Layer-3 Router. Each department is assigned a separate subnet for efficient communication and network management.

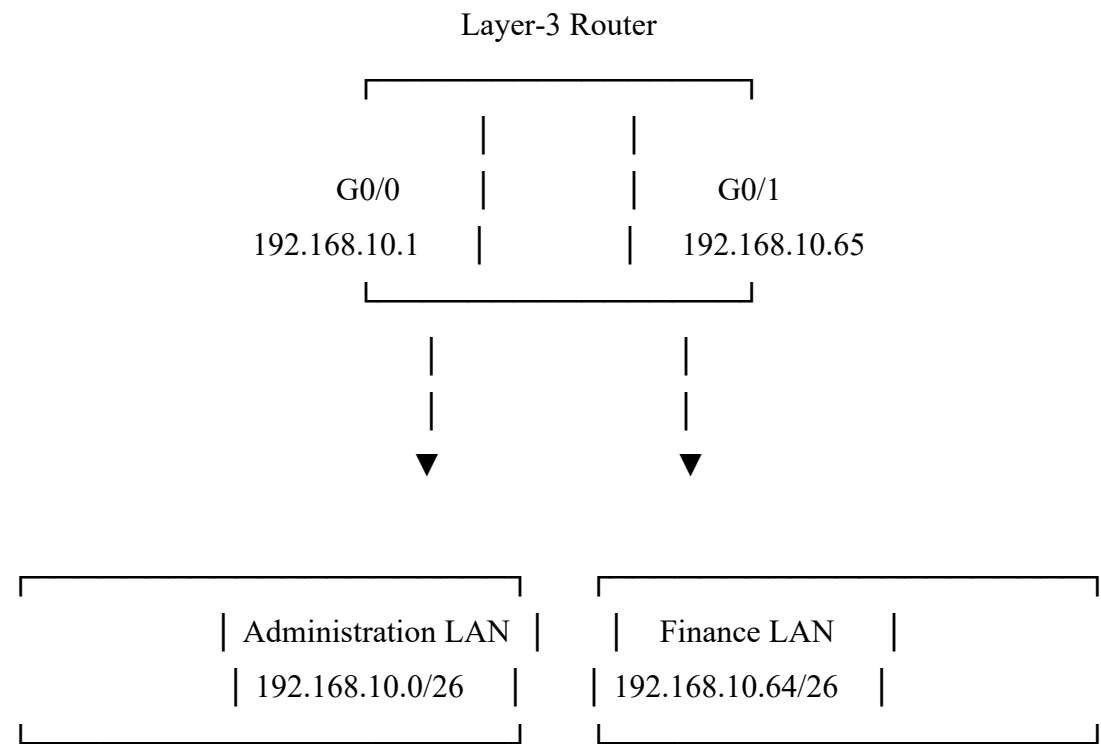
Given Information

Department	Network Address	CIDR Notation
Administration LAN	192.168.10.0	/26
Finance LAN	192.168.10.64	/26

Router Interface Configuration

Router Interface	IP Address
Administration Gateway	192.168.10.1
Finance Gateway	192.168.10.65

Network Topology



Task 1: Determine the Network Address and Broadcast Address

Explanation

For a /26 subnet:

- Subnet Mask = 255.255.255.192
- Total IP Addresses = 64
- Usable Host Addresses = 62

The network address is the first IP address in the subnet, and the broadcast address is the last IP address.

Administration LAN

- Network Address: 192.168.10.0
- Broadcast Address: 192.168.10.63

Finance LAN

- Network Address: 192.168.10.64
- Broadcast Address: 192.168.10.127

Answer

LAN	Network Address	Broadcast Address
Administration LAN	192.168.10.0	192.168.10.63
Finance LAN	192.168.10.64	192.168.10.127

Task 2: Identify the Valid Host IP Address Range

Explanation

The usable host IP addresses lie between the network address and the broadcast address.

- First Host = Network Address + 1
- Last Host = Broadcast Address - 1

Answer

LAN	First Host	Last Host
Administration LAN	192.168.10.1	192.168.10.62
Finance LAN	192.168.10.65	192.168.10.126

Task 3: Suggest Suitable IP Addresses for Three Computers

Explanation

The router interfaces already use:

- 192.168.10.1 for Administration
- 192.168.10.65 for Finance

Therefore, the computers should be assigned the next available IP addresses.

Administration LAN

Computer	Suggested IP Address
Computer 1	192.168.10.2
Computer 2	192.168.10.3
Computer 3	192.168.10.4

Finance LAN

Computer	Suggested IP Address
Computer 1	192.168.10.66
Computer 2	192.168.10.67
Computer 3	192.168.10.68

Task 4: Identify the Default Gateway

Explanation

A default gateway is the IP address of the router interface that hosts use to communicate with devices outside their local network.

- Devices in the Administration LAN use 192.168.10.1.
- Devices in the Finance LAN use 192.168.10.65.

Answer

LAN	Default Gateway
Administration LAN	192.168.10.1
Finance LAN	192.168.10.65

Subnet Layout

Administration LAN (/26)

Network Address : 192.168.10.0

First Host : 192.168.10.1 (Gateway)

PC-1 : 192.168.10.2

PC-2 : 192.168.10.3

PC-3 : 192.168.10.4

Last Host : 192.168.10.62
Broadcast Address : 192.168.10.63

Finance LAN (/26)

Network Address : 192.168.10.64
First Host : 192.168.10.65 (Gateway)
PC-1 : 192.168.10.66
PC-2 : 192.168.10.67
PC-3 : 192.168.10.68
Last Host : 192.168.10.126
Broadcast Address : 192.168.10.127

Conclusion

The two office departments have been successfully configured into separate /26 subnets using a Layer-3 router. Each LAN has its own network address, broadcast address, valid host range, and default gateway. The suggested IP addresses for the computers fall within their respective subnet ranges, ensuring proper communication within each LAN. The router gateways (192.168.10.1 for Administration and 192.168.10.65 for Finance) enable communication between the two departments and with external networks.

RUBRICS FOR CALCULATION

Scenario Based Assessment					
Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Scenario	20%	Demonstrates clear and complete understanding of the	Shows good understanding with minor gaps	Partial understanding; misses key aspects	Misinterprets or fails to understand the scenario

		scenario and context			
Identification of Key Issues	20%	Accurately identifies all relevant issues and factors involved	Identifies most key issues with minor omissions	Identifies limited issues; some irrelevant points	Unable to identify key issues
Application of Concepts	25%	Applies appropriate concepts effectively to address the scenario	Applies concepts with minor errors	Limited or partially correct application	Unable to apply relevant concepts
Decision Making	20%	Makes well-reasoned, logical decisions with strong rationale	Makes reasonable decisions with some justification	Makes weak decisions with limited justification	No clear decisions made or rationale provided
Clarity of Response	15%	Response is clear, structured, and logically presented	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand